



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to NAVANEE AGENCIES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

#### Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

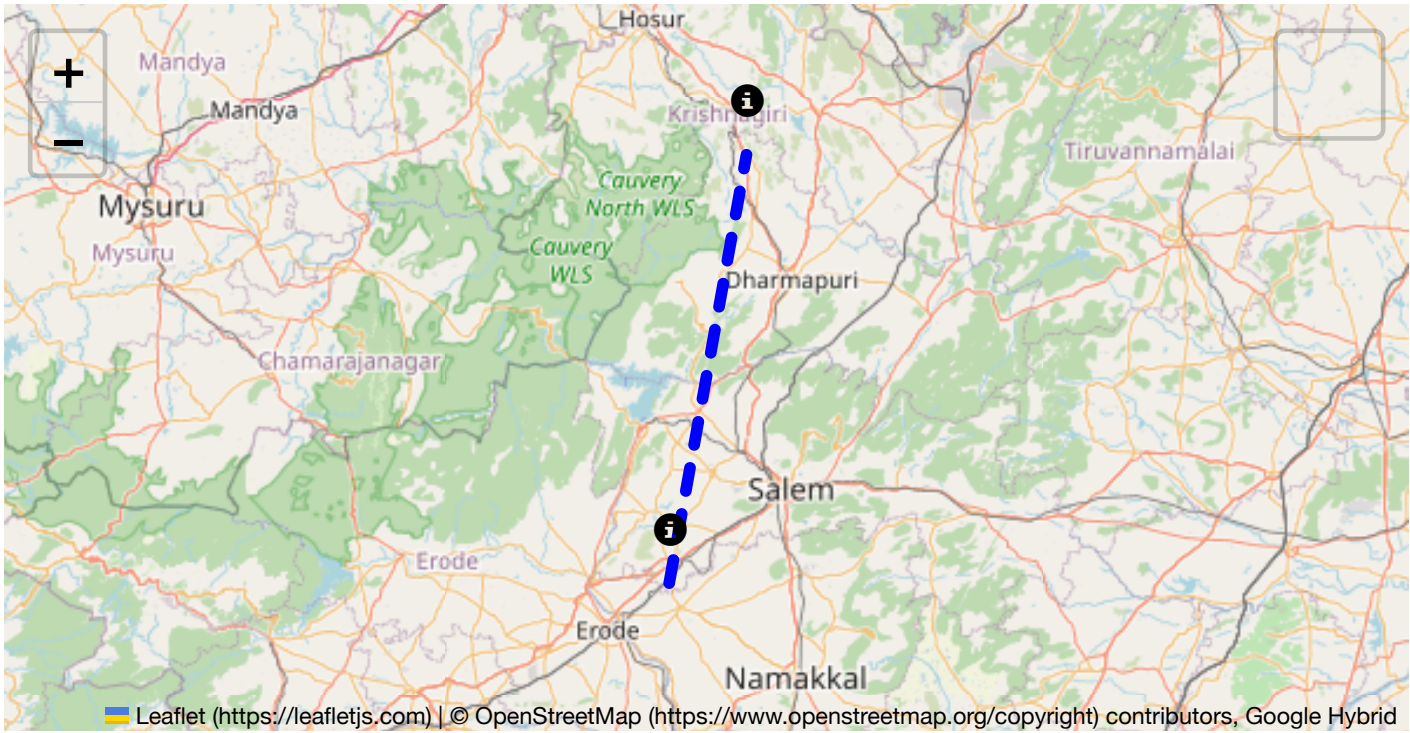
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

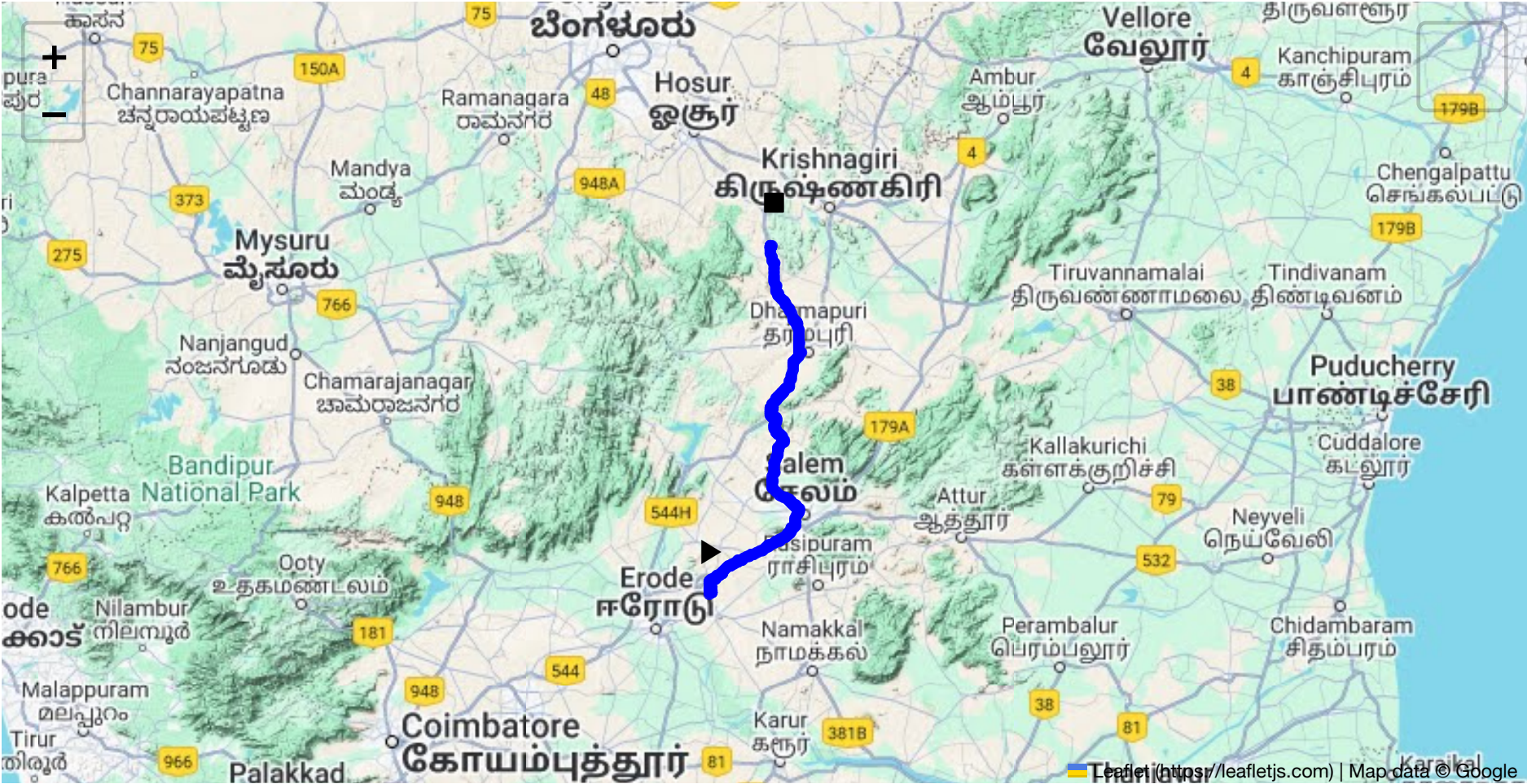
This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 140.18 km**  
**Estimated Duration: 2.5 hours**  
**Adjusted Duration (Heavy Vehicle): 3.1 hours**



Start: (11.4381, 77.8734)  
End: (12.42153, 78.04986)



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

- **Starting Point:** CVQF+23W, Sangagiri, Tamil Nadu 637301, India
- **Ending Point:** C3C2+G28, Sokkamatti Rd, Tamil Nadu 636805, India
- **Approximate Distance:** 140.18 kilometers
- **Estimated Travel Time:** Typically 2.51 hours for heavy vehicles with hazardous materials

This route generally involves traveling on national and state highways, known for their mixed traffic with both heavy vehicles and local transport.

### 2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather:** Tamil Nadu experiences tropical weather with hot, humid summers (March to May) and a monsoon season (June to September), followed by cooler winters (October to February).
- **Hazards:** Heavy rains during monsoon can lead to waterlogged roads and reduced visibility. High temperatures in summer may cause road fatigue and tire blowouts.

### 3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (7:00 AM - 9:00 AM) and evening (5:00 PM - 7:00 PM)
- **Congestion-Prone Areas:**
  - Sangagiri and Erode towns have higher congestion due to local commuter and commercial activity.
  - Intersections with major highways could also experience bottlenecks.

## 4. Assessment of Road Quality and Infrastructure

- **Highway Sections:** Generally well-maintained but may contain sections with potholes due to weather wear and tear.
- **Local Roads:** Narrower with limited shoulder space, which can be challenging for heavy vehicles.

## 5. Suggestions for Alternative Routes for Emergencies

- **Alternate Route:** Consider taking NH544 for a section if severe congestion or road closure occurs. It is slightly longer but often less congested.
- **Local Detours:** Utilize local state roads known to have less traffic to bypass specific congested areas.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

- Transportation of hazardous materials requires adherence to the Central Motor Vehicle Rules.
- Restriction zones typically include urban residential areas; plan to circumvent these as needed.

## 7. Overview of Historical Incidents

- Past incidents have involved overturning and spillage due to sharp turns and overcrowded road sections.
- Enhanced caution is advised in areas previously recorded for such incidents, especially during adverse weather.

## 8. Environmental Considerations and Sensitive Areas

- Areas near water bodies and agricultural zones along the route may have stricter environmental protection regulations.
- Special care should be taken to prevent contamination during transit.

## 9. Analysis of Communication Coverage

- Good coverage in major towns and along highways.
- Potential dead zones in rural sections; using GPS with offline maps recommended.

## 10. Estimated Emergency Response Times

- **Urban Sections:** Emergency response times typically between 15-30 minutes.
- **Rural Sections:** May extend to 45 minutes to 1 hour due to distance from major towns.

## 11. Overall Summary of Risk Assessment

This journey entails moderate risk, primarily due to traffic congestion and potential weather disruptions. Key risks include waterlogging during monsoon, high temperatures, and narrow local roads. Proper training on route-specific hazards, adherence to regulations, and consistent communication with dispatch should mitigate most risks. Emergency alternatives and support should be planned ahead, considering the nearest service centers and hospitals along the route. Regular risk assessments and incident reviews are advised to maintain high safety standards for transporting hazardous materials.



Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.45028, 77.87202 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.45309, 77.87325 | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.46400, 77.87323 | 30 KM/Hr    |
| 6  | Turn       | Medium     | 11.46765, 77.87037 | 30 KM/Hr    |
| 7  | Turn       | High       | 11.47403, 77.86841 | 15 KM/Hr    |
| 8  | Turn       | High       | 11.49149, 77.89601 | 15 KM/Hr    |
| 9  | Turn       | High       | 11.49224, 77.89601 | 15 KM/Hr    |
| 10 | Turn       | Medium     | 11.53501, 77.95707 | 30 KM/Hr    |
| 11 | Turn       | Medium     | 11.53601, 77.95856 | 30 KM/Hr    |
| 12 | Turn       | Medium     | 11.64180, 78.11808 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.64325, 78.11993 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.69519, 78.10775 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.73958, 78.05105 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.75835, 78.05016 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.87942, 78.08197 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.92585, 78.06507 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.93704, 78.05402 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.95818, 78.04978 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.95876, 78.04990 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.95988, 78.04948 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.96304, 78.05136 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.96399, 78.05135 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.96458, 78.05084 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.96496, 78.04988 | 30 KM/Hr    |
| 27 | Turn       | High       | 11.96580, 78.04923 | 15 KM/Hr    |
| 28 | Turn       | Medium     | 11.96675, 78.04939 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.96791, 78.05070 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 12.12058, 78.13283 | 30 KM/Hr    |
| 31 | Blind Spot | Blind Spot | 12.41986, 78.04639 | 10 KM/Hr    |
| 32 | Turn       | High       | 12.41674, 78.04642 | 15 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 33 | Turn      | Medium     | 12.41625, 78.04783 | 30 KM/Hr    |
| 34 | Turn      | High       | 12.41646, 78.04821 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                                                            | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 0  | hospital | Thilagam Hospital                                               | 11.4753573,<br>77.8685829 | 30 km/h     | Medium     |
| 1  | clinic   | Meyyazhagan Clinic                                              | 11.4752254,<br>77.8685671 | 30 km/h     | Medium     |
| 2  | hospital | Dr. Jaganathan Hospital                                         | 11.4752717,<br>77.8709312 | 30 km/h     | Medium     |
| 3  | hospital | Kathir Eye Hospital                                             | 11.4740976,<br>77.8681591 | 30 km/h     | Medium     |
| 4  | police   | Police Quarters                                                 | 11.4741243,<br>77.8673495 | 30 km/h     | Medium     |
| 5  | hospital | Sankari Hospital                                                | 11.4777496,<br>77.873892  | 30 km/h     | Medium     |
| 6  | clinic   | Dr. Dhanasekar Clinic                                           | 11.4793113,<br>77.8829247 | 30 km/h     | Medium     |
| 7  | hospital | Sri Vellaiyan Hospital                                          | 11.4798876,<br>77.8831394 | 30 km/h     | Medium     |
| 9  | hospital | C. N. Hospital                                                  | 11.55724, 78.0096         | 30 km/h     | Medium     |
| 10 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 11 | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 12 | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |
| 15 | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College          | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 19 | hospital | Bharathi Hospital, Salem                                        | 11.6492647,<br>78.1204131 | 30 km/h     | Medium     |
| 20 | hospital | Sisu Hospital Inida Ltd                                         | 11.6505276,<br>78.1205003 | 30 km/h     | Medium     |
| 21 | hospital | Aishwarya Hospital                                              | 11.6716291,<br>78.1267807 | 30 km/h     | Medium     |
| 22 | hospital | C.M. Hospital                                                   | 11.6747345,<br>78.1255034 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------|---------------------------|-------------|------------|
| 24 | pharmacy | Healthy Pharmacy                   | 11.6762714,<br>78.127744  | 30 km/h     | Medium     |
| 25 | pharmacy | Thulasi Pharmacy                   | 11.6761991,<br>78.1277377 | 30 km/h     | Medium     |
| 26 | hospital | Sri Gokulam Speciality Hospital    | 11.6788724,<br>78.1274258 | 30 km/h     | Medium     |
| 27 | hospital | Global Medical Center              | 11.6867049,<br>78.1186697 | 30 km/h     | Medium     |
| 28 | hospital | Manipal Hospital                   | 11.703072,<br>78.1026612  | 30 km/h     | Medium     |
| 31 | clinic   | Manimuthu Skin Clinic              | 11.7521189,<br>78.0484058 | 30 km/h     | Medium     |
| 32 | hospital | St. Joseph's Speciality Hospital   | 11.761611,<br>78.0522344  | 30 km/h     | Medium     |
| 33 | hospital | Anugraha Multi Speciality Hospital | 11.773549,<br>78.0546408  | 30 km/h     | Medium     |
| 34 | clinic   | JSA Clinic                         | 11.8700619,<br>78.0844982 | 30 km/h     | Medium     |
| 35 | hospital | L.R. Hospital                      | 11.8703354,<br>78.0844554 | 30 km/h     | Medium     |

### Crowded Spots

|    | type       | name                                                       | coordinates               | speed_limit | risk_level |
|----|------------|------------------------------------------------------------|---------------------------|-------------|------------|
| 13 | university | Vinayaka Missions Vinayaka Missions<br>Research Foundation | 11.5974056,<br>78.0843617 | 30 km/h     | Medium     |
| 14 | school     | Vinayaka Missions Annapoorani Matric<br>School             | 11.5971662,<br>78.0839876 | 30 km/h     | Medium     |
| 16 | school     | Dist Education Training                                    | 11.6133799,<br>78.1022409 | 30 km/h     | Medium     |
| 17 | school     | DIET - Salem                                               | 11.6144158,<br>78.1025078 | 30 km/h     | Medium     |
| 18 | school     | Diamond Rays Matric HR Sec School                          | 11.6385629,<br>78.1206259 | 30 km/h     | Medium     |
| 23 | college    | Sona College                                               | 11.6768,<br>78.1257052    | 30 km/h     | Medium     |
| 29 | university | Periyar University                                         | 11.7186758,<br>78.0771089 | 30 km/h     | Medium     |
| 30 | school     | Fathima Hr Sec School                                      | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |

|    | type    | name                              | coordinates              | speed_limit | risk_level |
|----|---------|-----------------------------------|--------------------------|-------------|------------|
| 36 | college | Laxminarayana Polytechnic College | 12.0421277,<br>78.103203 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.44029, 77.87544



Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.44900, 77.87406





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45028, 77.87202



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45309, 77.87325



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46400, 77.87323





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46765, 77.87037



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.47403, 77.86841



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49149, 77.89601





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49224, 77.89601



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53501, 77.95707



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53601, 77.95856





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64180, 78.11808



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64325, 78.11993



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.69519, 78.10775





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.73958, 78.05105



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.75835, 78.05016



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.87942, 78.08197





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.92585, 78.06507



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.93704, 78.05402



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95818, 78.04978





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95876, 78.04990



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95988, 78.04948



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96304, 78.05136





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96399, 78.05135



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96458, 78.05084



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96496, 78.04988





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96580, 78.04923



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96675, 78.04939



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96791, 78.05070





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.12058, 78.13283



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.41674, 78.04642

